

Homework #1: (Return by 22.9.25 in PDF -format)

An IC-circuit takes DC-current and sine wave shaped currents with amplitudes of 200 mA at frequencies of 25 MHz and $(10+n)$ MHz from a voltage rail (n is the last digit of your student ID number). Choose two bulk filter capacitors (<100 nF each) to make the supply voltage ripple as small as possible (**~ 10 mV peak-to-peak steady-state ripple voltage is sufficiently low**). The bulk capacitors available have an ESR of 10 mohm and ESL of 4 nH. Model bulk filter capacitors as the RLC-circuit ($R1$, $L2$, $C1$) in Fig. 1.

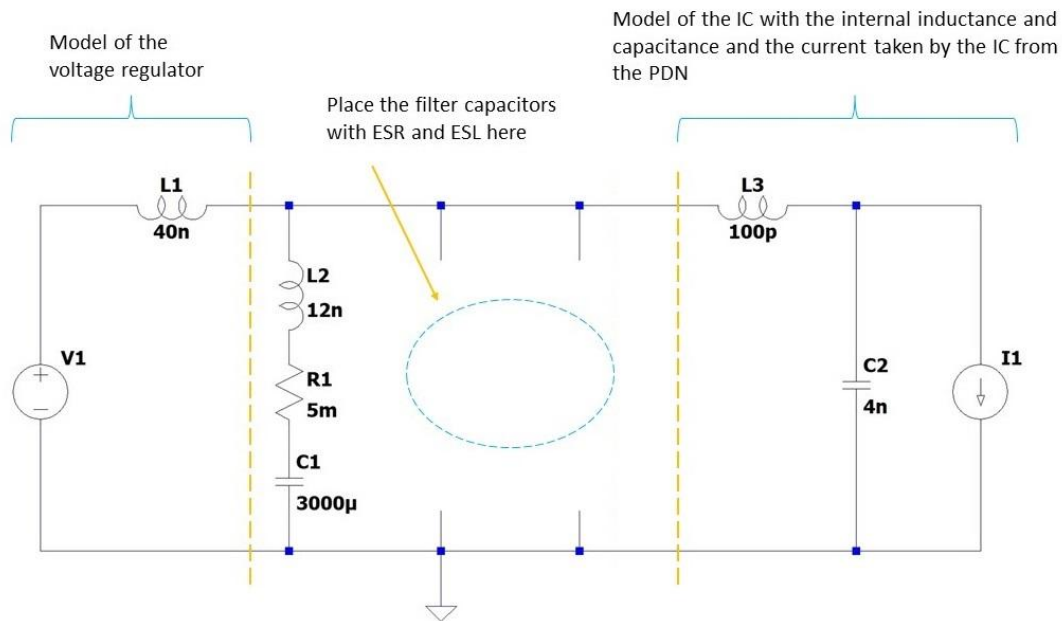


Fig. 1 Simulation model for the voltage regulator, filter capacitors and the IC.

In your answer, present an AC and a transient simulation result with and without filter capacitors. Choose a logarithmic y-scale in the AC simulation result.

In practice, how does the location of these bulk capacitors on the PCB affect the filtering frequencies (answer with a few sentences)? Help for the homework can be found in lecture material and if needed by email, jaakko.huikari@oulu.fi.

The simulations can be done with LTspice simulator found in <https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html>.

Quick guide to performing simulations with LTspice:

- Place a resistor, capacitor and a ground symbol in the schematic by pressing either R, C or G, correspondingly. Rotate components with CTRL-R.
- A current or voltage source can be placed from the menu under F2.
- The component values can be changed by clicking components with the right mouse button
- After clicking with the right mouse button, the current source type can be changed with the Advanced -button
- Wires can be placed with F3
- Press space for zoom to fit
- After selecting "Simulate -> Run" a voltage of a node can be probed by clicking a node in the schematic with the left mouse button
- In case you have any simulation related questions: jaakko.huikari@oulu.fi

Important practical notes on your return documents on this course:

- Name your return file in the following manner: ESD25 HW#n first name surname. As an example, in the case of homework #1 => ESD25 HW#1 Jaakko Huikari (first- /surname matching your name in the course's Moodle page (Enrolled users)).
- Return your documents in **PDF -format**. Do not return documents in Word- or other formats. Convert your word-document, for example, into PDF -format before you return it.
- Place your name and student ID number at the top edge of your return document. Use the name matching your name in the course's Moodle page (Enrolled users).
- Make sure your answers and simulation results are clearly visible in your return documents. If necessary, use cursors, zoom and screen captures to present sufficient graphic information (waveforms, values etc.). Make sure that figure axes are also visible in your return documents.
- Present the reasoning behind your answers/ results including theoretical basis and equations etc. (found in the lecture material). Presenting mere outcome is an inadequate answer.
- **Do not miss the deadline!**